

# YI DENG

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## EDUCATION

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### University of Virginia

Aug. 2025 – Dec. 2026

*M.S. in Computer Engineering* | GPA: 3.50 / 4.00

- Relevant coursework: Wireless Sensing, AI Hardware, GenAI.

### Shenzhen MSU-BIT University

Aug. 2021 – Jun. 2025

*B.Eng. in Electronic & Computer Engineering* | GPA: 3.71 / 4.00 (Top 30%)

- Core courses: Modern Communication Principles, Database Principles, Machine Learning.

## PUBLICATIONS

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### First-authored

- [1] **Y. Deng**, et al. "Cloud-Edge EEG Emotion Recognition under Communication Constraints," *IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, 2025.
- [2] **Y. Deng**, et al. "Reviving Affective Signals: A Generative Data Reconstruction Framework for EEG Digital Twins under Communication Constraints," *IEEE Open Journal of the Communications Society* (2026).
- [3] **Y. Deng**, et al. "The Influence of Mamba Model Hyper-parameters on Training Time and Accuracy," *IEEE International Conference on Smart Internet of Things (SmartIoT)*, 2024.

### Co-authored

- [4] M. Li, **Y. Deng**, et al. "Personalized Real-Time Epileptic Seizure Alert on Edge Devices via Few-Shot Meta-Learning," *IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, 2025.
- [5] M. Li, **Y. Deng**, et al. "GenDT-MAML: A Data-Efficient Framework for Personalized Seizure Prediction Using Generative Digital Twins and MAML," *IEEE Open Journal of the Communications Society* (2026).

## RESEARCH EXPERIENCE

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### Cloud-Edge EEG Emotion Recognition under Communication Constraints Nov. 2024 – Dec. 2025

First-authored: *CloudCom 2025, IEEE OJ-COMS (2026)*

- Designed and implemented an end-to-end EEG emotion recognition pipeline using CNN-LSTM-Attention architectures in PyTorch, processing multi-channel EEG signals from preprocessing through classification.
- Modeled cloud-edge deployment by simulating realistic communication constraints—packet loss, lossy compression, and bandwidth limits—to characterize how transmission degradation impacts downstream emotion classification accuracy.
- Developed generative reconstruction modules based on GAN and diffusion models to recover degraded EEG segments at the cloud side, restoring downstream inference accuracy under constrained channels.
- Conducted system-level analysis of accuracy, latency, and transmission cost trade-offs across deployment configurations, informing design guidelines for edge-deployed biosignal inference.

### Performance Analysis of Mamba State-Space Models

Apr. 2024 – Jul. 2024

First-authored: *SmartIoT 2024*

- Designed controlled experiments in a Linux environment to study how key hyperparameters (learning rate, batch size, SSM state expansion factor) affect Mamba training time and downstream accuracy.
- Built an automated benchmarking pipeline evaluating Mamba performance across multiple NLP datasets (SST-2, IMDB, AG News), enabling reproducible cross-dataset comparison.

- Identified performance trade-offs and training-stability regimes under different hyperparameter configurations, providing empirical guidance for deploying state-space models in resource-constrained settings.

## OTHER PROJECTS

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### Construction and Application of Raman Spectrometer

Apr. 2023 – May 2024

*Departmental research project, Shenzhen MSU-BIT University*

- Built and calibrated a Raman spectrometer and a UV-Vis absorption spectrometer from optical components, ensuring optical-path alignment and spectral accuracy.
- Designed and implemented additional measurement features, completing the engineering department's first in-house Raman spectrometer for educational and research use.

## INDUSTRY EXPERIENCE

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### Fusheng Innovation (Chengdu) Technology Co., Ltd.

Apr. 2024 – Sep. 2024

*Research and Development Engineer (Intern)*

- Designed and implemented backend modules in Go for a trading system, integrating broker APIs for order handling and status updates.
- Maintained relational databases for transactional data and supported testing of API behavior under network latency and failure scenarios.

### Shenzhen UBTECH Robotics Corp., Ltd.

Nov. 2024 – Feb. 2025

*Product Intern*

- Validated AI-based obstacle recognition through functional testing and supported system-level evaluation of robotic vacuum products, focusing on sensing, navigation, and edge cases.
- Collected and analyzed user feedback and test results to support iterative product improvements.

## TECHNICAL SKILLS

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- **Programming:** Python (proficient), C/C++, Go; experience with object-oriented design, algorithm implementation, and performance-aware development.
- **Machine Learning & Deep Learning:** PyTorch; CNN, LSTM, attention, and state-space (Mamba) architectures; generative models (GAN, diffusion); time-series modeling and feature extraction for biosignal learning tasks.
- **Signal Processing & Sensing:** EEG preprocessing and analysis; optical spectroscopy (Raman, UV-Vis); coursework exposure to WiFi CSI and FMCW radar pipelines for contactless physiological sensing.
- **Embedded & Control Systems:** MATLAB / Simulink for system modeling and simulation; basic controller design and tuning (e.g., PID).
- **Networking & Tools:** TCP/IP and HTTP communication, TCP/UDP module development in Python; Linux, SQLite, Git.

## HONORS & AWARDS

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- University-level Third-class Academic Scholarship, Shenzhen MSU-BIT University, 2022–2023 (Top 30%).
- Third Prize, 15th Blue Bridge Cup National Software and Information Technology Competition, University Group B (C/C++), Guangdong Province, Apr. 2024.